WILL YOU CAST YOUR NEXT VOTE ONLINE?

By Mike Talon

In light of recent events with our Presidential Election here in the US, many concerned citizens have been looking toward the Internet as a potential way to vote quickly, securely, and with less room for their votes to be misconstrued. The issues facing proponents of the plan are security risks, the "digital divide" cutting off the lower classes from the necessary technology, and devising a system that would guarantee that the person who's voting is who they say they are.

What may be somewhat surprising is that online voting has been successfully applied to some primary races during this election cycle. Arizona's Democratic Committee conducted their primary race online this year, with citizens logging into a secure website to cast their ballots. The results, however, were not without controversy. The Voter Integrity Project (www.voting-integrity.org) has already filed suit, alleging that the online ballot violates the Voting Rights Act of 1965.

Essentially, the issues surrounding Internet voting fall into two major categories: data security issues and physical security issues. Data security issues surrounding online votes are basically the same as those faced by financial institutions when trying to transmit and receive secure data over a very non-secure network, namely the Internet.

To start, whenever a voter sits down at their terminal to vote the data they receive from the polling place must be correct for their electoral district, with the correct candidates displayed in their correct order. Once the voter has decided who they will vote for, the data they send back to the polling place must be held with an incredible level of security.

Prying eyes must be kept away from the data stream so that the Constitutional rights to privacy are maintained. The data must be received by the polling place in exactly the same way as it was sent – without any corruption from accidental or deliberate sources. Finally, the computer systems at the polling place must take into account failed attempts to connect, broken connections in the middle of a transmission, and the end-user errors like clicking the "submit vote" button multiple times by accident.

In addition to these digital concerns, electoral districts must take into account the fact that many citizens won't have access to computers – or won't know where to get such access. Those who do will want some safeguard in place to ensure that they are the only person who can vote in their name, and that no voter can masquerade as someone else.

Dave Pell of Davenetics (www.davenetics.com) is one of the many online pundits who recommend a simpler idea for digital voting – kiosk systems in the polling place itself. This idea addresses the physical security issues, most notably how to ensure that the person voting is indeed the person they claim to be, and also most of the data-security issues.

First, a physical challenge such as a signature comparison can confirm voter identity – much like the current system in use in many electoral districts today. Once voter identity is confirmed, the voter enters the polling booth. Instead of the current punch-card system or other manual voting device, the voter is presented with an Internet kiosk, connected through a secure local network to an on-site polling server. The voter selects their candidates, and is even presented with a confirmation screen to ensure they are indeed voting for who they thought they were.

After voting is complete, the results are held locally until the polling place closes. At that time, data from each polling place can be securely sent via Virtual Private Networks or other secure systems to local, state, and federal tabulation offices for tallying. Since the votes are confirmed by the voter, and there are no physical checks (like chads, cards, or other physical media), tabulation is nearly instantaneous and non-ambiguous.

This idealized vision of electronic voting is not without it's problems. Kiosks would last only about 4-7 years if kept in good repair, as opposed to the decades that a manual voting apparatus can be kept in service. Also, data-security is still an issue when the votes are transmitted to the tabulation centers. Election officials will need to weigh these factors against

the benefits of the new systems before tossing out the existing apparatus.

At the current time, we may not be ready to fully convert our voting systems over to a digital format. Perhaps the gradual integration of polling-place digital systems with a future move to secure Internet voting is called for. This will help put an end to the possibility of another Election 2000 debacle, while still protecting the inalienable rights of the American Citizen to a private, secure and, most importantly, final vote.

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